

National University of Computer and Emerging Sciences

Lahore Campus

AI (AI2002)

Date: April 7th 2026

Course Instructors:

AA, RM, BR, UA, UU

MN, SI, ST, HW

23L-0771

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Roll No

BLS-6J

Section

Sessional-II Exam

Total Time (Hr): 1

Total Marks: 28

Total Questions: 3

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Student Name

Answer all questions on the provided sheet. No answer/rough sheets to be attached.
Show all steps where applicable.

CLO 1: Demonstrate understanding of AI based strategies to systematically search for a solution

Q1. A population of four chromosomes is given below. If random numbers are required, use them sequentially from the given list. Once a number is used, move to the next in order and do not repeat these numbers. [11 marks]

Random Number List: 0.82, 0.69, 0.57, 0.24, 0.62, 0.08, 0.79, 0.86, 0.26, 0.76, 0.43, 0.47, 0.89, 0.98, 0.58, 0.03, 0.57, 0.92

	C1	C2	C3	C4
Chromosome	11110000	10101010	00001111	01100111
Fitness	4	7	2	9

- a. Considering it's a maximization problem, which chromosomes will be selected as parents using fitness proportionate (roulette wheel) selection? Organize your working in tabular format with all steps. [3]

Chromosome#	Fitness	$F_i/\sum F_i$	Slices	First random number = 0.82	Second random number = 0.69
C1	4	$4/22 = 0.181$	0 - 0.181		
C2	7	$7/22 = 0.318$	0.181 - 0.499		
C3	2	$2/22 = 0.091$	0.499 - 0.59		
C4	9	$9/22 = 0.409$	0.59 - 1	Picked	Picked

Sum = 22

C4 gets picked twice

selected chromosomes = C4, C4 C4 gets picked twice

- b. Assume **C2** and **C4** are selected. Perform two-point crossover and show the resulting chromosomes. Determine crossover points using: Position = ceiling $[R \times N]$ where N = # of genes, R is random num. [3]

num. [3] $N = 8$ $R_1 = 0.57$ $R_2 = 0.24$

Position₁ = $\lceil 8 \times 0.57 \rceil$ Position₂ = $\lceil 8 \times 0.24 \rceil$

$= \lceil 4.56 \rceil$ $= \lceil 1.92 \rceil$

$= 5$ $= 2$

C2 → 10101010
C4 → 01100111

2 5

01 → 10 100 010
02 → 01 101 111

using Basic swap mutation crossover

- c. Assume chromosome **10010101** is selected for mutation. Perform two-bit mutation and show the resulting chromosome. Use the same formula $\lceil R \times N \rceil$ to determine mutation positions. [2]
- assuming that two bit mutation means to swap values at the 2 positions.

$N = 8$ $R_1 = 0.62$ $R_2 = 0.08$

Position₁ = $\lceil 8 \times 0.62 \rceil$ Position₂ = $\lceil 8 \times 0.08 \rceil$

$= \lceil 4.96 \rceil$ $= \lceil 0.64 \rceil$

$= 5$ $= 1$

10010101 → 00011101

- d. Why can a genetic algorithm converge prematurely to a local optimum? Suggest and briefly explain any two strategies to mitigate this issue. [3]

premature convergence can be caused due to lack of variety in the population.

Strategy 1 → increase the mutation rate, leads to greater variation.

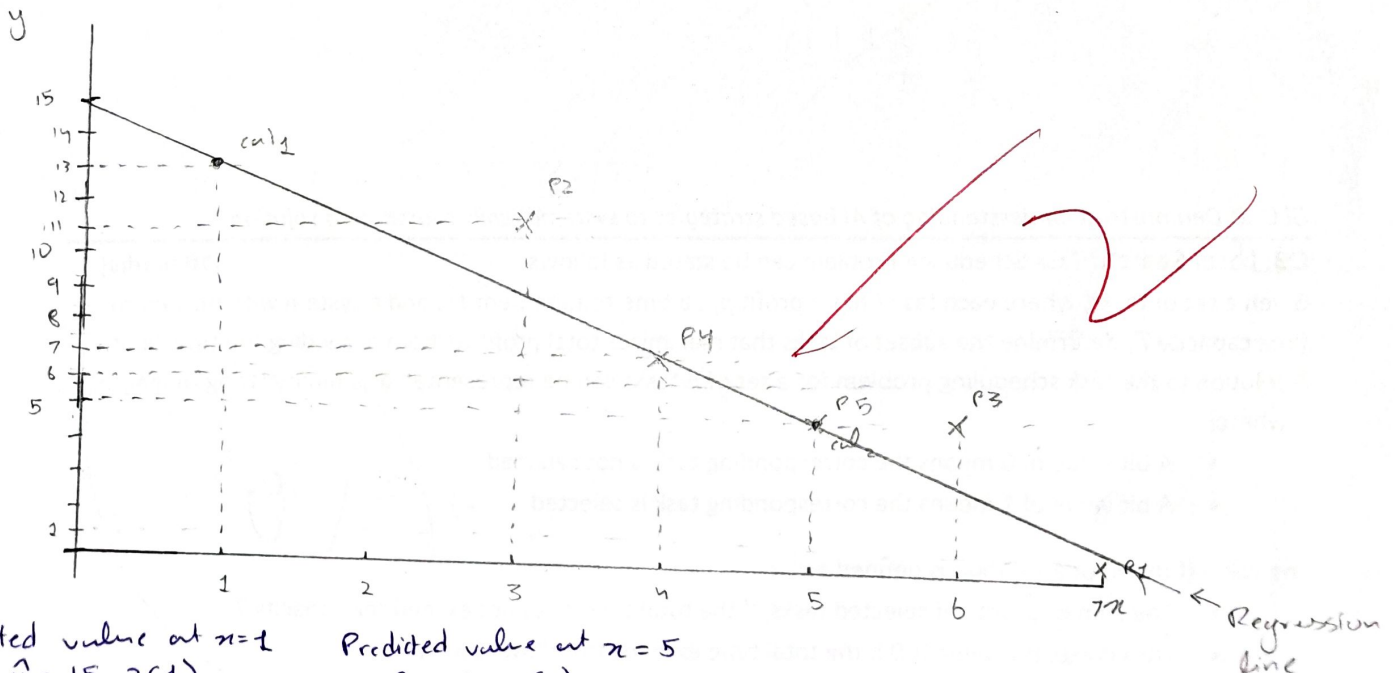
Strategy 2 → smaller percentage of elitism, smaller number of elites from previous generation moved to current generation, enables exploration.

CLO 3: Apply AI based techniques to solve real-world problems

Q2: A nutritionist is studying the relationship between the number of hours of sleep and the number of errors made in a cognitive test the next day. The following data was collected from 5 participants. [7 marks]

Participant	Hours of Sleep (X)	Number of Errors (Y)
✓ P1	7	1
✓ P2	3	11
✓ P3	6	5
✓ P4	4	7
✓ P5	5	5

- a. Plot the given data points and the regression line $\hat{y} = 15 - 2x$ on a graph. Clearly label axes.
To draw the regression line, compute any two points using the given equation. [2]



Predicted value at $x=1$
 $\hat{y} = 15 - 2(1)$
 $= 13$
 labelled calc 1
 Spring 2026

Predicted value at $x=5$
 $\hat{y} = 15 - 2(5)$
 $= 5$
 labelled calc 2
 Department of Computer Science

- b. Using the given model, compute predicted values and error/residual. Based on these errors, comment on the model's behavior and fit. [2+1]

x	y	$\hat{y}$	$(y - \hat{y})$
7	1	1	0
3	11	9	2
6	5	3	2
4	7	7	0
5	5	5	0
			$\Sigma(y - \hat{y}) = 4$

$$\hat{y} = 15 - 2x$$

total error = 4
The model follows the spread of data however the fit could be improved.

- c. Given that $R^2 = 0.82$ and $r = -0.91$, interpret both values in the context of this problem. [2]

R^2 is the coefficient of determination, shows how accurate the model is. 0.82 shows that it is fairly accurate.

r is the coefficient of correlation, the correlation between x and y is ~~-0.92~~ -0.91 negative.

CLO 1: Demonstrate understanding of AI based strategies to systematically search for a solution

Q3: Local Search: Task Scheduling Problem can be stated as follows:

[10 marks]

Given a set of tasks, where each task i has a profit p_i , a time requirement t_i , and a system with maximum time capacity T , determine the subset of tasks that maximizes total profit without exceeding the time limit. A solution to the task scheduling problem for a set of n tasks can be represented as a binary string of length n where:

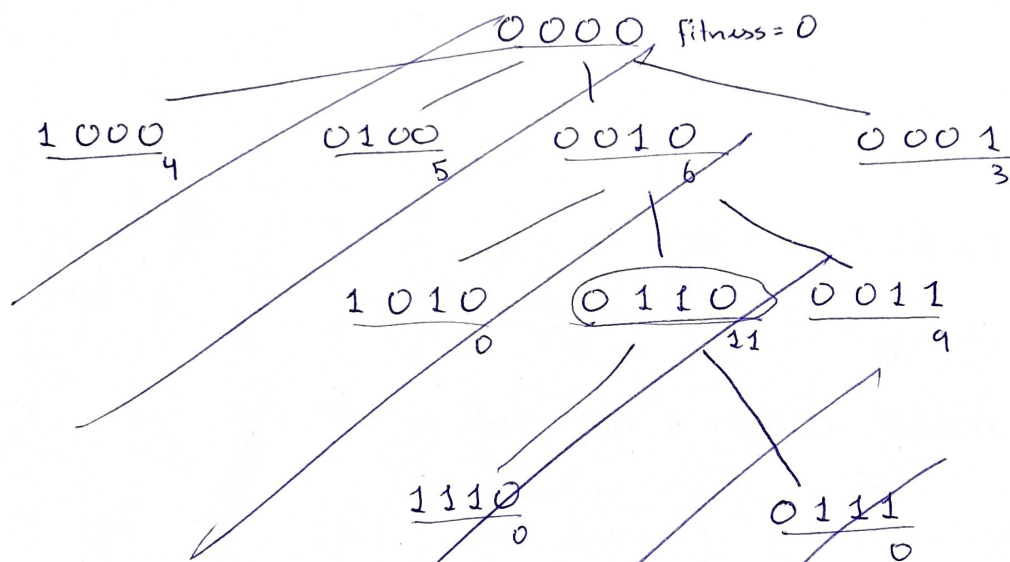
- A bit value of 0 means the corresponding task is not selected
- A bit value of 1 means the corresponding task is selected

The value (fitness) of a solution is defined as:

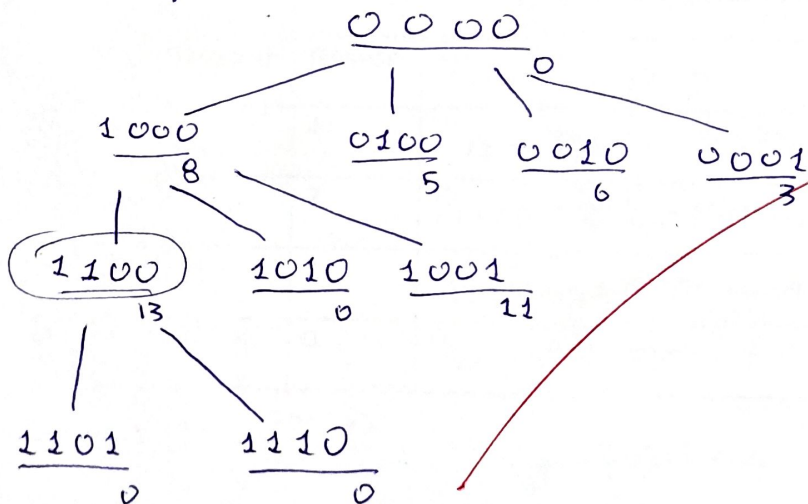
- The sum of profits of selected tasks, if the total time does not exceed the capacity T
- Otherwise, the value is 0 if the total time exceeds the allowed limit

Task Number	1	2	3	4
Profit	8	5	6	3
Time	4	3	5	2

For the set of four tasks shown above and a system with maximum time capacity 8, start from the initial solution 0000, where each bit represents whether a task is selected or not selected. Apply the hill-climbing (local search) algorithm to find the best solution. Assume that the successors/neighbours of a solution are created by setting exactly one of the zero bits to 1. You must show all intermediate solutions generated in each iteration. The process should stop when no neighbouring solution yields a better fitness value than the current solution.
max time capacity = 8



The final solution is 0110



our final solution is 1100
it has no more optimal neighbors.